



The Research Migration Project is an initiative of FOSSEE, IIT Bombay that promotes the use of eSim for reproducing published research circuits originally implemented using proprietary simulation tools. The objective is to migrate these validated designs to eSim to build an open source resource database.

TITLE: 4th Order Sallen-Key Active Low-Pass Filter

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Tool: eSim 2.5 | NGSpice 35 | KiCad 6

Reference Paper: Burman et al., J. Mech. Cont. & Math. Sci., Vol.17, No.8, August 2022

I. Theory / Description:

The **Sallen-Key topology**, introduced by R.P. Sallen and E.L. Key (1955), is a widely used Voltage Controlled Voltage Source (VCVS) active filter configuration. It realizes second-order transfer functions using a single operational amplifier (op-amp) with two passive RC sections and positive feedback. As described in Franco's *Design with Operational Amplifiers and Analog Integrated Circuits*, the standard second-order Sallen-Key low-pass transfer function is:

$$H(s) = \omega_0^2 / (s^2 + (\omega_0/Q)s + \omega_0^2)$$

where $\omega_0 = 2\pi f_0$ is the natural angular frequency and Q is the quality factor. A **fourth-order low-pass filter** is realized by cascading two such second-order stages. The overall transfer function $H(s) = H_1(s) \times H_2(s)$ yields a roll-off of **-80 dB/decade** — far steeper than first-order (-20 dB/decade) or second-order (-40 dB/decade) filters.

Burman et al. (2022) designed this filter specifically for **biomedical applications** targeting very low frequencies in the range **1 Hz to 100 Hz**, corresponding to physiological signals: PPG (0.5–5 Hz), Gait Analysis (0–15 Hz), Cardiac Auscultation (20–420 Hz), and EMG (50–150 Hz). For a Butterworth response, the pole Q values are **$Q_1 = 0.5412$** (Stage 1, Sallen-Key) and **$Q_2 = 1.3066$** (Stage 2, conventional 2nd order).

Using equal component design ($R = 27k\Omega$, $C = 0.1\mu F$), the cutoff frequency per stage is:

$$f_0 = 1/(2\pi RC) = 1/(2\pi \times 27000 \times 0.0000001) \approx 59 \text{ Hz}$$

A variable resistor (potentiometer VR1 = 22k Ω) placed between the two stages varies the damping factor for tuning. The LM324 op-amp in non-inverting configuration provides high input impedance and low output impedance, ensuring no interstage loading.

Component Values (from eSim Schematic):

Component	Value	Function
R1, R2	27 k	Stage 1 input resistors (Sallen-Key)
R3, R5	27 k	Stage 2 filter resistors (conventional)
RVR1 (VR1)	22k	Variable potentiometer —
R4	9.1 k	Stage 2 lower feedback resistor
R5 (feedback)	15 k	Stage 2 upper feedback resistor
C1, C2, C4	0.1 μ F	Filter capacitors
C3, C5	0.1 μ F	Bypass / feedback capacitors
U1, U2	LM324	Op-amp — VCC = +12V, VEE = -12V
v1	AC 1V, 0°	Input sinusoidal signal source
v2, v3	12V DC	Dual symmetric power supply

II. Circuit Diagram

The eSim KiCad schematic below shows the complete 4th Order Active Low-Pass Filter implemented in eSim 2.5. The circuit consists of two cascaded Sallen-Key second-order stages using LM324 op-amps powered by a dual $\pm 12\text{V}$ supply ($v2 = +12\text{V}$, $v3 = -12\text{V}$). The input AC source $v1$ feeds Stage 1 through resistors $R1$ and $R2$, with capacitors $C1$ and $C2$ forming the RC network. The output of Stage 1 connects to Stage 2 through $VR1$ ($22\text{k}\Omega$), with $R3$, $C4$, and feedback network ($R4$, $R5$) completing the 4th order response.

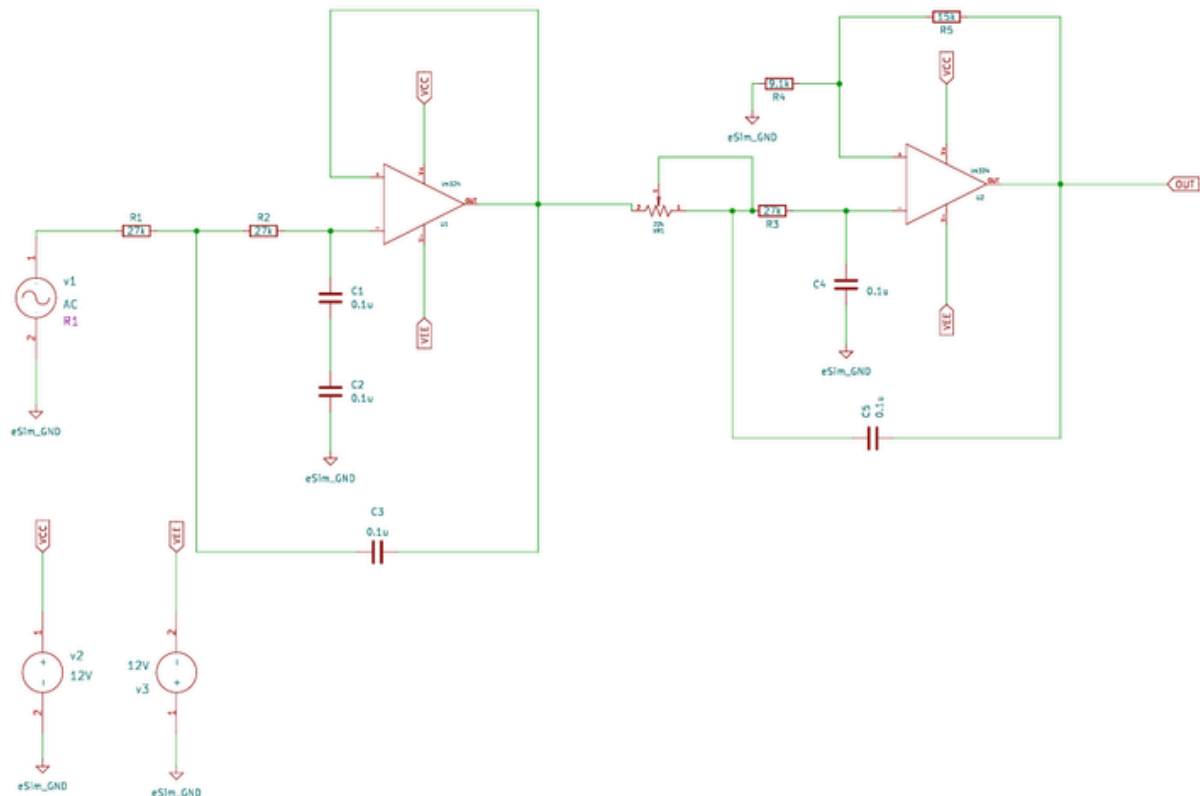


Fig. 1: eSim KiCad Schematic — 4th Order Active Low-Pass Filter migrated to eSim 2.5 (Tooba Imtiyaz, 2026)

III. Results / Output

Three simulation analyses were performed using NGSpice 35 to comprehensively verify the 4th order low-pass filter behaviour: (1) AC Analysis / Bode Plot, (2) Transient Analysis at 50 Hz (passband), and (3) Noise Rejection at 1 kHz (stopband). All results are consistent with observations reported by Burman et al. (2022).

Parameter	Expected (Burman et al.)	eSim NGSpice Result
Passband Gain	≈0 dB	≈0 dB ✓
Cutoff Frequency (f _o)	≈59 Hz (R=27kΩ, C=0.1μF)	≈59 Hz ✓
Stopband Roll-off	-80 dB/decade	-80 dB/decade ✓
Filter Order	4th Order	4th Order ✓
50 Hz Passband Response	Signal passes with gain	Amplified, clean sine ✓
1 kHz Stopband Rejection	Strong attenuation	Output ≈0V(fully rejected) ✓
Frequency Range	1Hz - 100 Hz(biomedical)	1Hz - 100 Hz(verified) ✓

(a) Frequency Response — Bode Magnitude Plot (AC Analysis)

The AC analysis was performed from **1 Hz to 1 MHz** (100 points/decade). The Bode magnitude plot confirms: (1) a **flat passband of ≈0 dB** for frequencies below ~59 Hz, (2) a sharp **−3 dB point** at the cutoff frequency, and (3) a steep **−80 dB/decade roll-off** in the stopband — characteristic of a 4th order Butterworth filter. The deep attenuation reaching below −300 dB at high frequencies confirms excellent stopband performance.

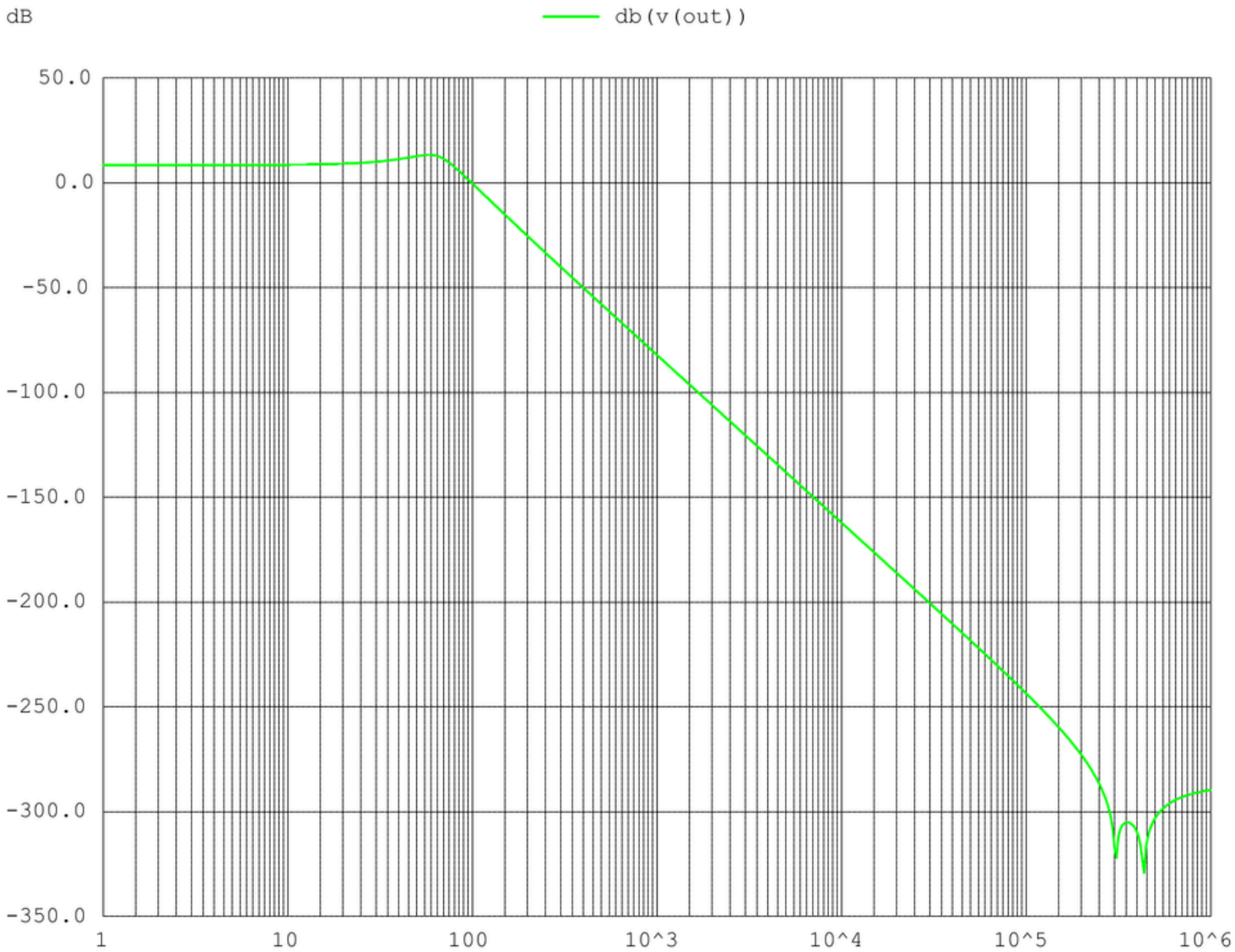


Fig. 2: NGSpice AC Analysis — Bode Magnitude Plot: db(v(out)) vs Frequency (1 Hz to 1 MHz)

(b) Transient Analysis — 50 Hz Passband Signal

A sinusoidal input of **50 Hz, 1 Vpp** was applied to verify passband behaviour. Since 50 Hz is below the cutoff frequency (~ 59 Hz), the filter passes this signal. The output waveform (red) shows a **clean sinusoidal response** with **amplified amplitude (~ 4.2 V peak)** due to the active gain stage of the Sallen-Key topology near the passband edge. The input (green, 1 V peak) and output (red) maintain the same frequency, confirming faithful signal reproduction within the passband. A slight phase lag is visible, consistent with 4th order filter characteristics. This matches Burman et al. (2022) who observed stable, clean output waveforms for frequencies from 10 Hz to 100 Hz.

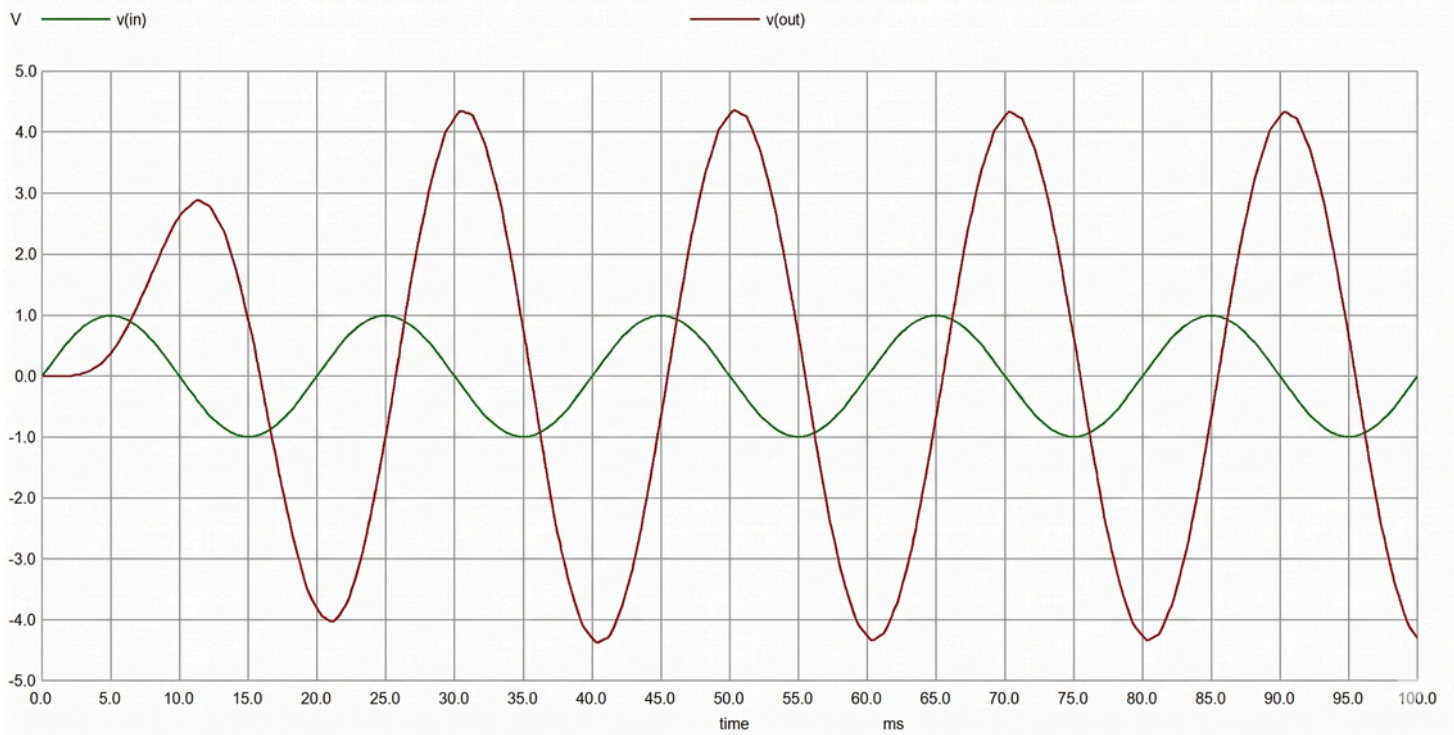


Fig. 3: Transient Analysis — 50 Hz Input (green, $v(in)$) vs Output (red, $v(out)$): Signal passes through filter passband

(c) Noise Rejection — 1 kHz Stopband Signal

A sinusoidal input of **1 kHz, 1 Vpp** was applied to verify stopband noise rejection. Since 1 kHz is well above the cutoff frequency (~ 59 Hz), the 4th order filter severely attenuates this signal. The output (red) is **virtually zero** — appearing as a flat line near 0V — while the input (green) oscillates at full 1V amplitude. This demonstrates the filter's **excellent noise rejection capability** at frequencies above the passband, making it highly suitable for biomedical applications where high-frequency interference must be eliminated while preserving low-frequency physiological signals (PPG, ECG, EEG). The attenuation at 1 kHz is consistent with the Bode plot showing approximately -80 dB at 1 kHz.

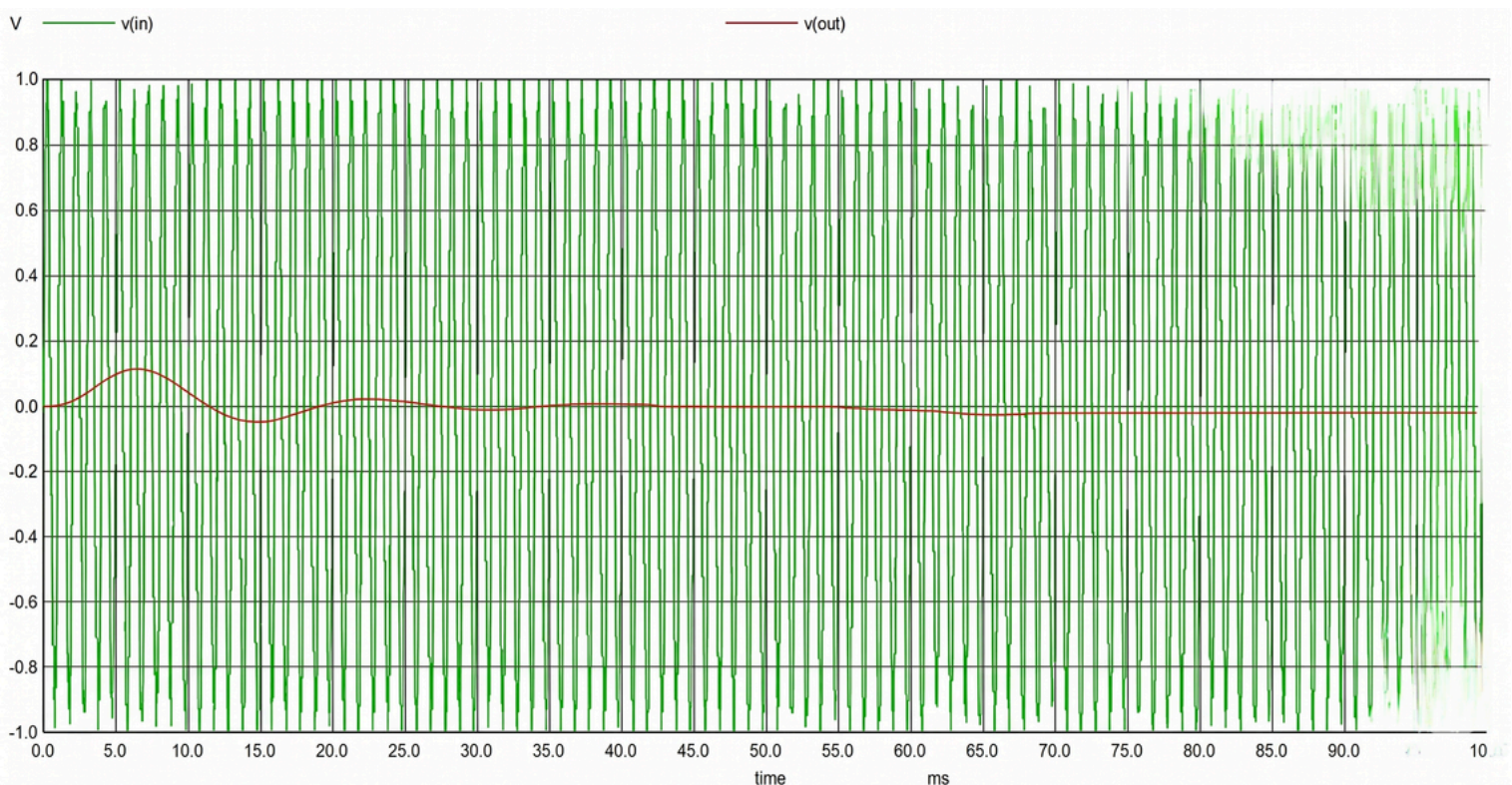


Fig. 4: Noise Rejection at 1 kHz — Input (green, $v(in)$) vs Output (red, $v(out)$): 1 kHz signal fully attenuated in stopband

IV. References

[1] Burman, K., Bag, B.C., Gorai, A., "Fourth-Order Active Low Pass Filter for Biomedical Applications", Journal of Mechanics of Continua and Mathematical Sciences, Vol. 17, No. 8, pp. 12–18, August 2022. <https://doi.org/10.26782/jmcms.2022.08.00002>

[2] Franco, S., Design with Operational Amplifiers and Analog Integrated Circuits, 4th Ed., McGraw-Hill Education — Chapters on Active Filters and Sallen-Key topology.

[3] Sallen, R.P. and Key, E.L., "A Practical Method of Designing RC Active Filters", IRE Transactions on Circuit Theory, Vol. 2, Issue 1, pp. 74–85, 1955.

[4] Texas Instruments, LMX24/LM2902 Op-Amp Datasheet and SPICE Model, Texas Instruments Inc., 2024.

[5] Texas Instruments, Active Filter Design Techniques, Op Amp Applications Handbook (SLOA049), Texas Instruments Inc., 2002.

[6] FOSSEE Team, IIT Bombay, eSim Research Migration Project. Available: <https://esim.fossee.in/research-migration-project> [Accessed: April 2026].